# MSG No. 136.497 $P4_2/m'nm$ [ Type III, tetragonal ]

Table 1: Wyckoff site: `2a`, site symmetry: `m'.mm`

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ 0,\ 0]$ | `[1,4,7,8,11,12,13,16]` |
| 2 | $\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$ | `[2,3,5,6,9,10,14,15]` |

Table 2: Wyckoff site: `2b`, site symmetry: `m'.mm`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0,\ 0,\ \frac{1}{2}\right]$ | `[1,4,7,8,11,12,13,16]` |
| 2 | $\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$ | `[2,3,5,6,9,10,14,15]` |

Table 3: Wyckoff site: `4c`, site symmetry: `2/m'..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0,\ \frac{1}{2},\ 0\right]$ | `[1,4,13,16]` |
| 2 | $\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$ | `[2,3,14,15]` |
| 3 | $\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$ | `[5,6,9,10]` |
| 4 | $\left[\frac{1}{2},\ 0,\ 0\right]$ | `[7,8,11,12]` |

Table 4: Wyckoff site: `4d`, site symmetry: `-4'..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$ | `[1,4,14,15]` |
| 2 | $\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$ | `[2,3,13,16]` |
| 3 | $\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$ | `[5,6,11,12]` |
| 4 | $\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$ | `[7,8,9,10]` |

Table 5: Wyckoff site: `4e`, site symmetry: `2.mm`

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ 0,\ z]$ | `[1,4,7,8]` |
| 2 | $\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$ | `[2,3,5,6]` |
| 3 | $\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$ | `[9,10,14,15]` |
| 4 | $[0,\ 0,\ -z]$ | `[11,12,13,16]` |

Table 6: Wyckoff site: `4f`, site symmetry: `m'.2'm`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ x,\ 0]$ | `[1,8,11,16]` |
| 2 | $\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{2}\right]$ | `[2,5,10,15]` |
| 3 | $\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$ | `[3,6,9,14]` |

*continued ...*

Table 6

| No. | position | mapping |
|---|---|---|
| 4 | $[-x, -x, 0]$ | `[4,7,12,13]` |

Table 7: Wyckoff site: `4g`, site symmetry: `m'.m2'`

| No. | position | mapping |
|---|---|---|
| 1 | $[x, -x, 0]$ | `[1,7,12,16]` |
| 2 | $\left[x+\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}\right]$ | `[2,6,9,15]` |
| 3 | $\left[\frac{1}{2}-x, \frac{1}{2}-x, \frac{1}{2}\right]$ | `[3,5,10,14]` |
| 4 | $[-x, x, 0]$ | `[4,8,11,13]` |

Table 8: Wyckoff site: `8h`, site symmetry: `2..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0, \frac{1}{2}, z\right]$ | `[1,4]` |
| 2 | $\left[0, \frac{1}{2}, z+\frac{1}{2}\right]$ | `[2,3]` |
| 3 | $\left[\frac{1}{2}, 0, z+\frac{1}{2}\right]$ | `[5,6]` |
| 4 | $\left[\frac{1}{2}, 0, z\right]$ | `[7,8]` |
| 5 | $\left[\frac{1}{2}, 0, \frac{1}{2}-z\right]$ | `[9,10]` |
| 6 | $\left[\frac{1}{2}, 0, -z\right]$ | `[11,12]` |
| 7 | $\left[0, \frac{1}{2}, -z\right]$ | `[13,16]` |
| 8 | $\left[0, \frac{1}{2}, \frac{1}{2}-z\right]$ | `[14,15]` |

Table 9: Wyckoff site: `8i`, site symmetry: `m'..`

| No. | position | mapping |
|---|---|---|
| 1 | $[x, y, 0]$ | `[1,16]` |
| 2 | $\left[\frac{1}{2}-y, x+\frac{1}{2}, \frac{1}{2}\right]$ | `[2,15]` |
| 3 | $\left[y+\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}\right]$ | `[3,14]` |
| 4 | $[-x, -y, 0]$ | `[4,13]` |
| 5 | $\left[\frac{1}{2}-x, y+\frac{1}{2}, \frac{1}{2}\right]$ | `[5,10]` |
| 6 | $\left[x+\frac{1}{2}, \frac{1}{2}-y, \frac{1}{2}\right]$ | `[6,9]` |
| 7 | $[-y, -x, 0]$ | `[7,12]` |
| 8 | $[y, x, 0]$ | `[8,11]` |

Table 10: Wyckoff site: `8j`, site symmetry: `..m`

| No. | position | mapping |
|---|---|---|
| 1 | $[x, x, z]$ | `[1,8]` |
| 2 | $\left[\frac{1}{2}-x, x+\frac{1}{2}, z+\frac{1}{2}\right]$ | `[2,5]` |
| 3 | $\left[x+\frac{1}{2}, \frac{1}{2}-x, z+\frac{1}{2}\right]$ | `[3,6]` |
| 4 | $[-x, -x, z]$ | `[4,7]` |
| 5 | $\left[x+\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}-z\right]$ | `[9,14]` |

*continued ...*

Table 10

| No. | position | mapping |
|---|---|---|
| 6 | $\left[\frac{1}{2}-x,\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$ | [10,15] |
| 7 | $[x,\, x,\, -z]$ | [11,16] |
| 8 | $[-x,\, -x,\, -z]$ | [12,13] |

Table 11: Wyckoff site: 16k, site symmetry: 1

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\, y,\, z]$ | [1] |
| 2 | $\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$ | [2] |
| 3 | $\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$ | [3] |
| 4 | $[-x,\, -y,\, z]$ | [4] |
| 5 | $\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$ | [5] |
| 6 | $\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$ | [6] |
| 7 | $[-y,\, -x,\, z]$ | [7] |
| 8 | $[y,\, x,\, z]$ | [8] |
| 9 | $\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$ | [9] |
| 10 | $\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$ | [10] |
| 11 | $[y,\, x,\, -z]$ | [11] |
| 12 | $[-y,\, -x,\, -z]$ | [12] |
| 13 | $[-x,\, -y,\, -z]$ | [13] |
| 14 | $\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{2}-z\right]$ | [14] |
| 15 | $\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$ | [15] |
| 16 | $[x,\, y,\, -z]$ | [16] |